Kevin Phan

Sr. Software Engineer | Full-Stack Development

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SKILLS

- **Languages:** JavaScript, Python, PHP, Java, Go, C#, Ruby, C/C++, SQL
- **Front-end:** HTML, CSS, JavaScript/TypeScript, React/Vue, Next.js, Redux, Context API, Bootstrap, Material UI, Tailwind CSS, D3.js, ApexCharts.js, Quill.js, Popper.js, Rechart.js, GSAP
- **Back-end:** Node.js/Express/Nest.js, Python/Django/FastAPI, PHP/Laravel/CI, Java/Spring MVC/Spring Boot, Go, Rails, C/C++, SQL, Redis, RESTful APIs, GraphQL, gRPC, OAuth, JWT, Kafka, RabbitMQ
- **Database:** MySQL, PostgreSQL, MongoDB, DynamoDB
- **Cloud/DevOps:** Github Actions, Docker, Kubernetes, AWS, GCP, Azure, Terraform, Jenkins
- **Testing:** Unit testing (Jest, Pytest, JUnit, PHPUnit, Google Test), E2E testing (Cypress, Playwright, Selenium)
- **Architecture:** Monorepo, Microservices, Atomic Design
- **Other:** WebRTC, Webpack, Agile, SEO, Figma, Sketch, Adobe, A/B testing, React Query, Vite, ELK, Prometheus, Grafana, Git, Github, Gitlab, Bitbucket, Jira

PROFESSIONAL EXPERIENCE

AMAZON • Senior Software Engineer — Feb 2019 - Feb 2025

❖ **Frontend**

- Developed real-time inventory sync dashboards using **Next.js**, **React Query**, and **GraphQL**, helping sellers monitor sync health, identify errors, and drill down into item-level updates.
- Built the Inventory by Location Tool using **React** and **React Query**, enabling sellers to filter, search, and update inventory at specific store locations in bulk.
- Created the Eligibility Admin Portal for internal teams to manage seller eligibility using **Vite**, **Material UI**, and **React** with RBAC (Role-Based Access Control).
- Leveraged **Chart.js**, **Victory.js**, and **D3.js** to create dynamic data visualizations, providing insights into inventory, order synchronization, and fulfillment performance metrics.
- Optimized frontend **SEO** using server-side rendering (SSR) with **Next.js**, enhancing search engine indexing and improving organic rankings.
- Built dynamic and interactive UI components using **React** and **Redux**, ensuring state management was optimized and components were reusable and maintainable.
- Implemented **GraphQL + Apollo Client** on the frontend for efficient data fetching with cache-first strategies and optimistic UI updates.
- Optimized frontend performance by implementing **code splitting**, **lazy loading**, and **image optimization**, resulting in a 40% improvement in page load time and enhanced user experience.
- Implemented automated testing with **Jest** and **Cypress**, ensuring high-quality and bug-free code through continuous integration.

❖ **Backend & Microservices**

- Developed a distributed sync engine using **Python** and **Go** to automate real-time inventory updates from seller systems to Amazon's catalog, with logic via **SQS** and **DynamoDB** Streams.
- Built onboarding APIs in **Node.js** and **Java** for store and item registration, integrated with **PostgreSQL**.
- Implemented audit logging and secure access workflows using **Amazon KMS**, **Cognito**, and **AWS IAM** to ensure compliance in internal services.
- Designed and implemented a secure, scalable API Gateway using **AWS API Gateway** and **Kong**, efficiently routing frontend requests to appropriate backend microservices.
- Designed, optimized, and managed scalable data solutions and high-throughput queries with **PostgreSQL**, **MongoDB**,

DynamoDB, **Elasticsearch**, and **Firebase**, ensuring data consistency and reliability for critical financial services.

- Optimized backend data retrieval by integrating **Redis** for caching, resulting in improvements in API response times.
- Implemented comprehensive logging and monitoring for all backend services using the **ELK Stack**, **AWS CloudWatch**, and **Datadog**, allowing the team to proactively monitor system health and resolve issues.
- Managed scalable microservices with **Docker** and **Kubernetes** on **AWS EKS**, enabling auto-scaling, and high availability.
- Integrated with **GitHub Actions** for automated CI/CD workflows, reducing release time and improving system reliability.

❖ **Collaboration, & Mentorship**

- Led cross-functional squads in the delivery of high-impact features, driving alignment between engineering, product, and design.
- Collaborated with DevOps teams to manage infrastructure using **Terraform** and **AWS** services (**Lambda**, **S3**, **EC2**, **ECR**, **ECS**), reducing operational costs while ensuring high availability.
- Mentored junior engineers through pair programming, code reviews, and onboarding programs, accelerating team productivity.

Doxy.me • Full Stack Developer — Jul 2014 - Jan 2019

- Contributed to the development of a real-time telemedicine platform, enabling secure peer-to-peer virtual consultations between healthcare providers and patients.
- Engineered core platform features to support live video consultations and real-time diagnostics, improving remote patient care and clinical decision-making.
- Integrated HL7 and DICOM standards for interoperability with EHRs, PACS systems, and third-party healthcare applications, ensuring regulatory compliance and seamless data exchange.
- Led performance optimization and system scalability efforts, enhancing responsiveness, uptime, and throughput while maintaining HIPAA compliance and secure data handling.
- Collaborated with clinicians, UX designers, and product managers to optimize user workflows and ensure platform accessibility in compliance with FDA and WCAG guidelines.

EDUCATION

University of California, Santa Cruz — Aug 2010 - May 2014

Bachelor of Computer Science